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- 4A GPSE

Ball regulation

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Contents

List of Figures

1 Introduction

A fascinating technical challenge at the intersection of several engineering disciplines such as mechanics, electronics, control systems, and computer science is controlling the position of a moving object on an inclinable platform. The project goal is the conceptualization and the realization of a real-time control system to keep a ball in the center of an inclinable platform. The main objective is to stabilize the ball by varying the inclination of the platform according to the location of the ball by means of modern technologies like real-time image processing, physical modeling, and a PID (proportional, integral, and derivative) control system.

The ball position is detected using a camera and analyzed by an image information algorithm. Corrections are made to the inclining surface using small devices that were fed from the ball position data processed in real-time by the image processing algorithm. This type of system is not only a good illustration of the intertwining of various technologies, but it is also a platform that allows learning about advanced concepts in automatic control, computer vision, and dynamic modeling.

A Raspberry Pi 5 along with a Pi V2.1 camera fits the performances stated so far by the project with execution speed and detection precision. The Raspberry Pi 5 stands as the best for handling the simultaneous image capture, actual data processing, and executing the control algorithm due to its improved power and multitasking capability. The Pi V2.1 camera provides different resolution and frame rates so that a ball can be tracked with sufficient precision required by the system.

This technical report detail the design, implementation, and testing phases of the system, documenting technology choices, challenges met, and solutions developed. It also presents the results derived, through different methods, through essentially examining the performance of the system in terms of accuracy, responsiveness, and robustness.

The objective was to look into how modern technologies can be integrated to solve a problem and that gives an opportunity for further development and change. This work expresses multidisciplinarity, where the cooperation of different engineering backgrounds is assured for ambitious goals and pushes the limits of what can be accomplished from the technical point of view.

2 Usage Scenario

This section presents a detailed usage scenario for the control of a ball project, illustrating the system's functionality and practical applications. The scenario outlines the interaction between key components, emphasizing the constraints and requirements to achieve precise control and stability. By simulating real-world conditions, this scenario provides a clear context for the system's operation.

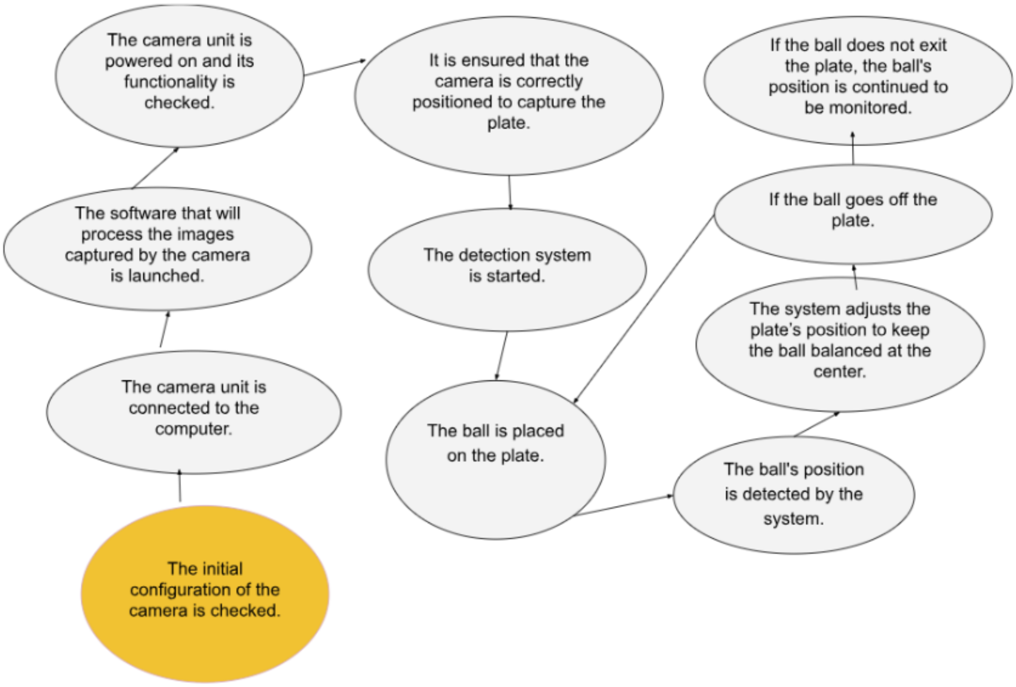


Figure 1: Usage Scenario, source: Salma Nahrou.

This ball control project revolves around a cooperative interaction between several modules, namely, a motorized platform, a camera dedicated to image analysis, and a PID control algorithm to track the ball stably and accurately on the platform, taking into account various constraints on inertia, ball speed, or motor performance. It takes place in an operational framework where the system detects the position of the ball in its dynamics relative to the platform on which it rests, tilts the platform using servo motors, and performs calculations that allow it to respond in real-time to the movement of the ball and maintain a position of equilibrium. By illustrating the project with examples of real-life situations such as sudden changes in the ball's position or rapid rotations of the platform, this project demonstrates how the latter can maintain perfect balance while reacting to disturbances. This introduction thus provides an overview, presenting the project's relevance and the technical solutions implemented to achieve its objectives.

3 Functional Synopsis

3.1 Fast diagram

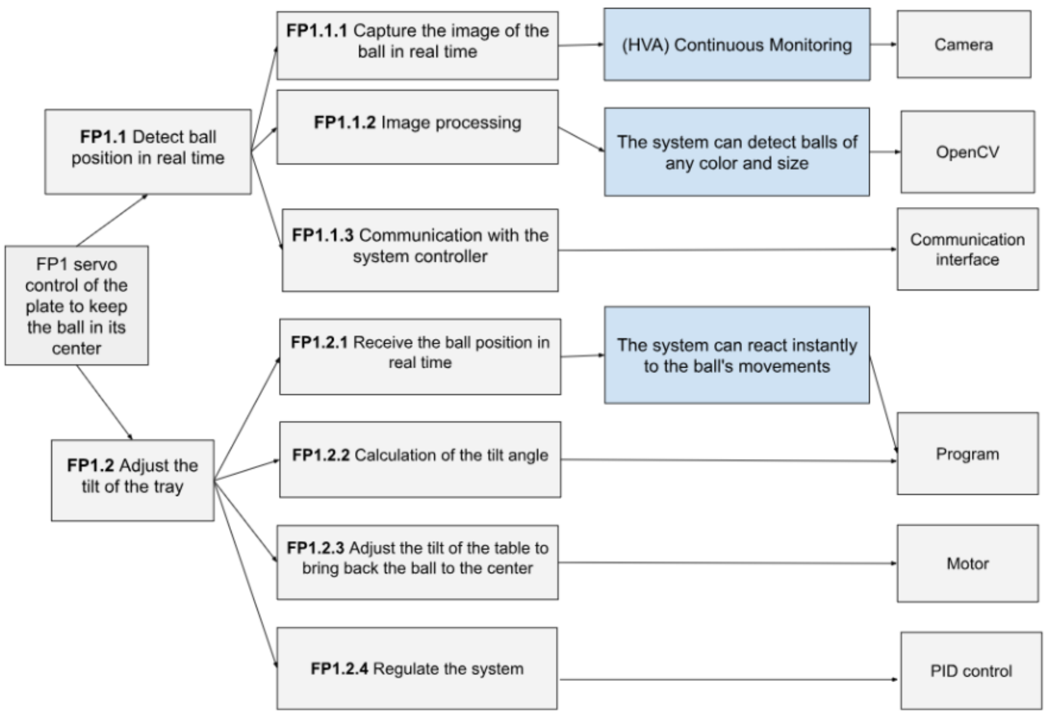


Figure 2: Fast Diagram, source: Farhi Omar .

The ball control system is based on two main functions: detection of the ball position and tilting of the tray. The first function uses cameras to take real-time images of the tray, which are processed in conjunction with the OpenCV library to extract the coordinates of the ball through segmentation and tracking algorithms. These data are then transferred to the microcontroller via a relatively fast communication interface to minimize latency. The second function is the tilt of the tray in order to bring the ball back to the center. A PID algorithm will continuously compute the necessary angles to correct the ball's position and motors will even more accurately tilt the tray into the desired position. The entire system works closed loop so as to provide a very fast response and be highly stable in the event of an external disturbance.

3.2 Constraints table

Constraints	Description
Maximum ball speed	v²=2.a.d=2.g×sin(10°).0.3 v=0.45 m/s
Detection in movement	The camera must detect the ball even when the tray rotates.
Frame rate (FPS)	The camera must have FPS > 20 to minimize response delay and track movements in real time
Detection precision	The center of the ball must be located with a precision of 3 mm.
Speed of PID	PID calculations must be fast enough to ensure system stability in real time.
Mechanical Precision	The motors must be able to keep the tray stable, even with the load of the moving ball.
Motor response time	The motors must react quickly and follow the ball's position changes

Figure 3: Constraints functions, source: Salma Nahrou

With certain limitations on the system,it is designed to work to perfection. The ball is limited at maximum speeds of 0.45 m/s, which requires good control over the ball to keep up with stability reaction.The camera is required to capture and treat the ball during tray rotations, in which case it should provide more than 20 FPS to avoid delays in real-time tracking. Detection accuracy is highly rated, which calls for finding the center of the ball within a 3 mm margin.The PID controller needs to react almost instantly to maintain stability.With that, because of the ball's weight,the motors should provide a stable reaction in control to maintain the tray position and to react quickly to any change of position to ensure accurate and reliable operations.

4 System Architecture

4.1 Application

The primary objective of this system is to dynamically stabilize a ball on a tilting platform by continuously tracking its position and adjusting the platform's angle in real time. This is achieved using a combination of computer vision, PID control algorithms, and precise servo motor actuation.
The system is designed to:

- Track and detect the ball's position accurately using a camera module.

- Process input data in real time to compute necessary adjustments to the platform's tilt angle.

- Control platform adjustments via a servo motor to ensure the ball remains centered on the platform.

4.2 Capella

The Arcadia Method (Architecture Analysis and Design Integrated Approach) is a broader MBSE (Model-Based System Engineering) methodology.It supports the Capella tool by providing a structured framework to define, analyze, and validate system architectures. Arcadia focuses on understanding stakeholder needs, managing complexity, and aligning system design with operational requirements while enabling iterative development.

The first step of this method involves carrying out the System Analysis using a Mission Capabilities Blank Diagram (MCB), where the actors interacting with the system and the capabilities or functions the system "must be able to do" to achieve a specific objective are defined. For this diagram, the process began with listing the actors belonging to the system's environment and interacting with it. These include the user, the environment, the ball, the light, and the stakeholder Polytech Orleans. Next, the roles of these actors must be defined. The user and the environment can initiate the system by making the ball move. The ball must be returned to the center of the tray whenever it deviates. Lastly, the light must illuminate the tray to minimize the ball's shadow .

The second step involves describing the system architecture by developing both the functions of the system's actors and the functions within the system using a System Architecture Diagram (SAB). This diagram mainly consists of transforming the capabilities identified in the previous step into functions, except for elements like the light, the ball, or the stakeholder, where an associated function within the system must be created. The lighting reduces the shadow of the ball on the camera, improving detection accuracy. The project's goal is to keep the ball centered on the tray, while the stakeholder aims to have a functional system to showcase at open doors to promote the school.

Finally, the last step of the Arcadia method describes all system components using the Physical Architecture Diagram (PAB) to address the previously identified functions from the SAB. This step requires detailing every aspect of the system, its components, their functions, and how they are interconnected. First, the camera continuously captures images, and a program processes the video stream to detect the ball in real-time. This program runs on a microcomputer that controls the entire system. Once detected, the program registers the ball's center coordinates, comparing them to the target position—the center of the tray. A PID controller then calculates the angular values needed to adjust the motors. If the ball moves away from the center, the system ensures it is brought back. The angular values generated by the PID controller are converted into voltage signals sent to the motors, which adjust the tilt of the tray until the ball stabilizes at the center. All these processes are monitored by the microcomputer. The stakeholder, Polytech Orleans, requires this final step to be operational to present a functional prototype at open doors, thereby promoting the school.

5 Technical Choices and Perspectives

The system is composed of a tray holded by mechanical supports. A PID control is applied on the motors monitoring those supports because we need the ball to remain in the center of the tray and with access to the ball's position obtained through a camera.

5.1 Image processing

5.1.1 Introduction

The Image processing part is very important because the entire system relies on it. The camera films in real-time the tray and the ball. This part allows us to calculate the error which is the entry of the PID control. This means that we need a clean detection of the ball in order to see its trajectory continuously despite any motion blur. This section aims to develop the objectives and constraints, then the technical description, the tests, our conclusions, as well as the next steps.

5.1.2 Objectives and constraints

Objectives

The image processing primary objectives are:

- Detect the ball in real-time.
- Detect the tray to find its center's coordinates.
- Ensure the detection despite the color and the size of the ball.
- Execution of the program per frame (delay) < 30ms.

Constraints

The image processing constraints are:

- The ball can go as fast as 0.45 m/s.
- The camera must detect the ball when the tray rotates and moves it up and down.
- The center of the ball must be detected as precise as 3mm.

5.1.3 Usage scenario

The following figure illustrates the operating steps of the image processing program.

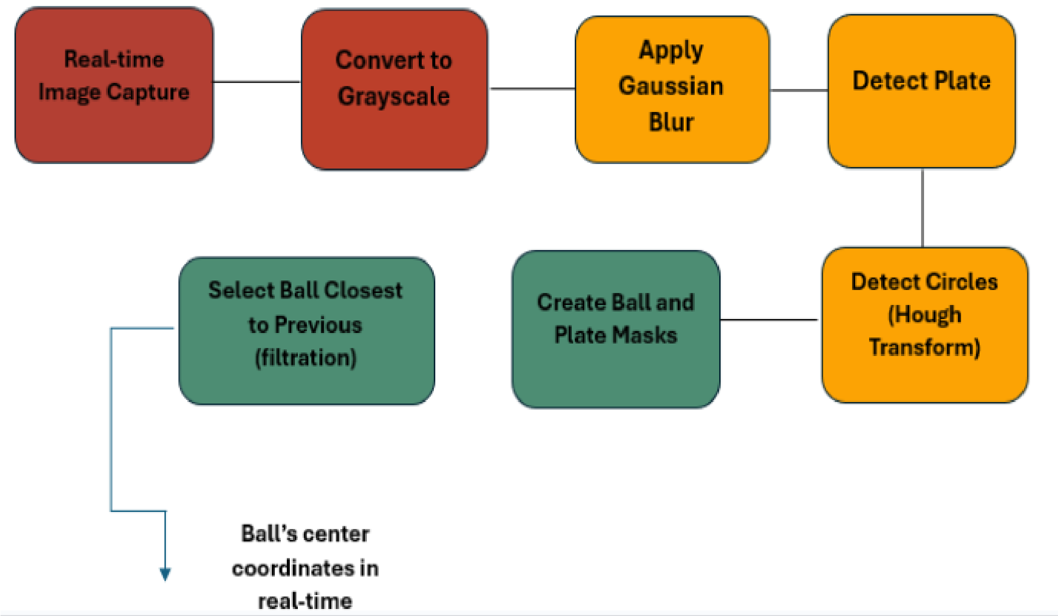


Figure 4: Diagram of the Image processing section

When the program is launched, the camera begins capturing images in real time. These frames are first converted to grayscale and then processed using a Gaussian blur to reduce noise. The plate and ball are both circular, so the same circle detection method (Hough Transform) is applied to identify them.

To optimize performance, the plate is detected only once at the beginning. The program runs a loop that searches for the tray and registers its center coordinates. These fixed coordinates are considered the target position for both X and Y axes.

Once the plate is detected, the system switches to a second infinite loop that continuously detects the ball in the video stream. From all detected circles, the algorithm selects the one closest to the previously detected position to ensure stability. Ball and plate masks are also created to assist with isolation.

The ball's center coordinates are then calculated in real time and compared to the plate's center. These two values serve as inputs to the PID control system, which calculates the error to correct the ball's position. The process continues indefinitely until the user presses the "Q" key to quit the program.

5.1.4 Technical description

Working process To generate the image processing code, First, it was necessary to study what this code is supposed to do. According to the usage scenario, The video stream of the camera has to be started to then detect the tray and the ball. The main languages able to do this were Python and C++ but Python is way more easy to learn, with more resources and easier to implement. Moreover, the programming language is Python because then it is easier to implement the entire code together since the other sections like calculation and PID control are coded in Python as well. Then, The choice had to be made between the Hough transform or the HSV (Hue Saturation Value) filtering methods to detect the ball but The HSV filtering method couldn't be made to work. Indeed, after trying the HSV filtering method and the find Contours function, It could be seen on the camera that it was working great when. Only the ball was detected, but there was a significant framerate drop when trying to detect both the tray and the ball at the same time. In addition, it was not possible to make a first loop to only detect the tray. The program would launch, but the detection of the ball was unconvincing. After looking up the issues, it became clear that this method is suitable for segmenting objects by color and shape, but is limited in detecting complex shapes like perfect circles, especially with multiple detection loops. To optimize the code with the Hough transform method, the parameters of the Hough transform had to be adjusted. The most important ones are the inverse ratio of the accumulator dp, the Canny edge detector threshold param1, and the accumulator threshold param2. The first parameter, dp, equals one when the accumulator has the same resolution as the frame and equals two when the accumulator has half the resolution. Using dp as one is less efficient because it is slower but increases precision, whereas dp equals two is faster and more efficient for detecting the ball in motion. For the next parameter, param1 corresponds to the upper threshold used by the Canny Edge Detection algorithm to detect edges in the image. Setting a low value for param1 helps detect more edges, including noise, which can increase the number of false circles detected. To address this, a blur mask had to be implemented. There was an issue with the blur mask initially reducing the camera's frames per second by ten, but it was eventually implemented without reducing the framerate. This mask is crucial because the Gaussian blur smooths the image and enhances geometric shapes, reducing noise in the video stream, which is vital for more precise ball contour detection with param1. Without this blur, many false circles appeared, complicating optimal ball detection. Finally, param2 defines the minimum number of votes required for a geometric shape to be considered a circle. This value must be high enough to avoid detecting false circles. Ultimately, detecting the tray first and registering its center coordinates, followed by real-time tracking of the ball, was considered to reduce the Raspberry Pi 5's consumption and increase the frame rate. The final result was approximately 30 frames per second on the camera.

Technical details

After comparing several cameras, it was realized that the camera used by the previous group working on this project was more than sufficient to serve as input for the image processing. Reusing a camera purchased by the school for this project is an excellent initiative, as it not only reduces expenses but also limits the environmental impact, contributing to a more sustainable and responsible approach. The model chosen is the Raspberry Pi V2.1. This camera is Pi-compatible, which is ideal since the Raspberry Pi 5 was selected to manage the entire system. Below is a detailed technical description compared to another camera.

Characteristic	**Pi Camera V2 (8 MP)**	**Arducam 5 MP**
Resolution	8 MP (3280 × 2464 pixels)	5 MP (2592 × 1944 pixels)
Frame Rate	60 fps (720p), 30 fps (1080p)	30 fps (720p), 15 fps (1080p)
Interface	CSI	CSI et SPI
Advantages	Higher resolution, excellent compatibility with Raspberry Pi	High frame rate, cost-effective
Disadvantages	Less smooth for fast-moving objects	Lower resolution

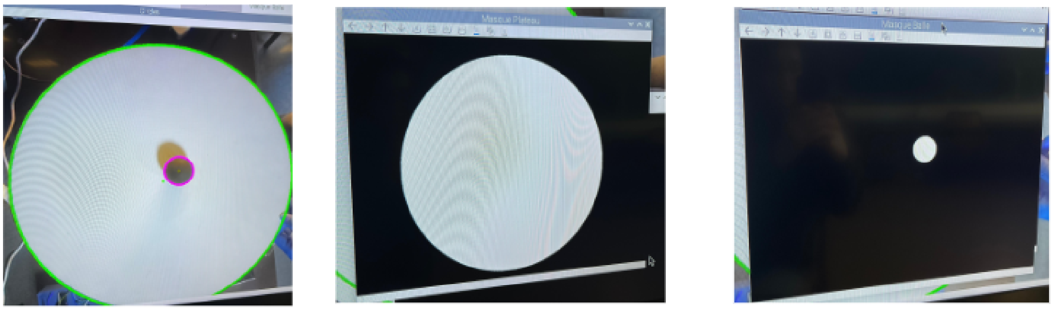
Table 1: Comparison of Pi Camera V2 and Arducam 5 MP

Not only the Pi V2.1 is good for the CSR approach but it is also an excellent choice because of its good resolution, it has a high frame rate and a lot of advantages like a better light sensor that captures less noise, better image quality, and it is more compatible with the Pi 5. The ball is not moving fast enough for this camera to be considered a disadvantage.

5.1.5 Testing

In this part, different methods will be implemented in the program to determine its effectiveness. First, the execution time per frame will be calculated. The goal is to measure the delay between the video stream input and the position of the ball in the output. This is crucial because real-time reactions of the motors are required through the PID control, which demands the delay to be less than 30 ms. To achieve this, the execution time of the most important functions in the code needs to be evaluated. The code lines implemented are detailed in the Image processing program. After executing the code, pressing 'Q' to stop the program results in an average time per frame of t = 0.0287s, corresponding to the delay. Comparing this value with the one indicated in the objectives shows that the results are indeed below the 30ms delay limit. This confirms that the program respects the constraints and performance objectives. Additionally, the average number of frames per second varies between 33 and 25, which is more than sufficient, even when the HSV (Hue Saturation Value) code doesn't meet the performance, objectives.To summarize, the program meets all technical performance constraints but is not yet ready for use in the global system due to the inadequate detection of the ball.

The following images illustrate the visual output of our image processing algorithm for detecting the circular plate and the ball:



Circle Détection Plate Mask Ball Mask

Figure 5: Image Processing – Test Results

1-Raw detection result: The green circle represents the detected edge of the plate, while the circle shows the detected ball. When the plate is nearly flat, both are accurately detected.

2-Plate mask only: This image shows the binary mask applied to isolate the circular plate. It is correctly segmented when the lighting and angle are appropriate.

3-Ball mask only: A clean binary mask showing the detected ball. This step is used to track the ball independently from the background or plate surface.

These results demonstrate that the detection is reliable in static or near-horizontal conditions. However, as the plate tilts or lighting conditions vary, the contours may become distorted, which reduces accuracy. This highlights the importance of robust contour detection and potential improvements through camera calibration and lighting control.

5.1.6 Conclusion

During the initial testing phase, it was observed that the ball was not consistently detected when using a white ball on a white tray, despite several optimization attempts. This limitation appears to be mainly due to the lack of contrast and the presence of shadows cast by the ball. To address this, several improvements were explored, including enhancing lighting conditions and increasing contrast between the tray and the ball. Comparative tests using orange and black balls showed significant improvement in detection accuracy. The orange ball was detected more effectively than the white one, though the tracking was still not perfectly fluid. The black ball, however, yielded the most reliable detection. Before integrating the image processing with the PID control system, this limitation must be addressed. Potential solutions include painting the tray in a darker color to improve contrast or adapting the algorithm to use color-based detection. Additionally, using a camera with a higher frame rate could improve tracking fluidity.

5.2 Physical modeling

Introduction

The focus is on the global movement of the ball because it is not only necessary to compare its position with the center of the tray but also to analyze its trajectory, as the dynamics of the ball are complex. The x and y axes are represented on the table and not on the tray; otherwise, the movement of the ball when the tray rotates would not be considered.

5.2.1 Forces Acting on the Ball

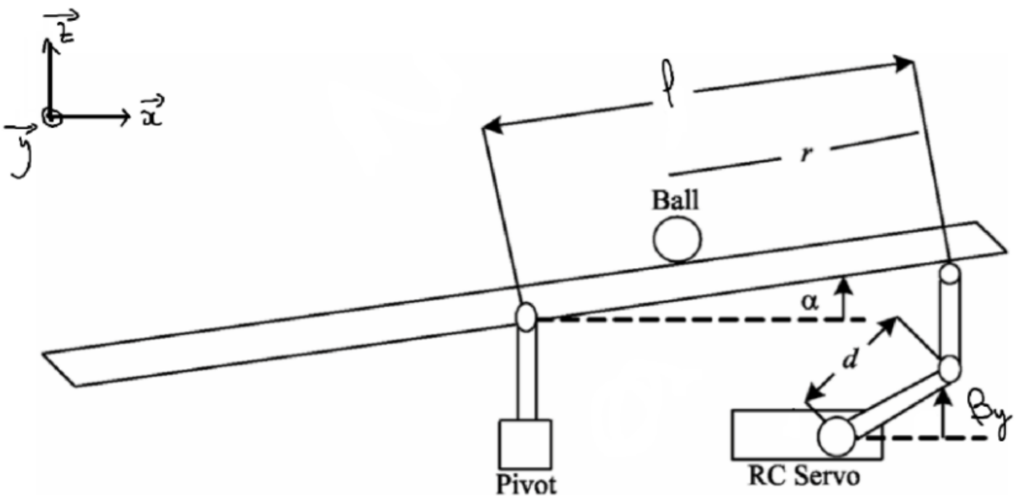


Figure 6: Schematic representation of our physical modeling

The translation force is the force of gravity acting on the ball to move it in a specific direction on the board. The inertial force depends on the mass of the ball and its acceleration or deceleration during its movement. These two forces are used to model and stabilize the movement of the ball on an inclined surface, taking into account its dynamics.

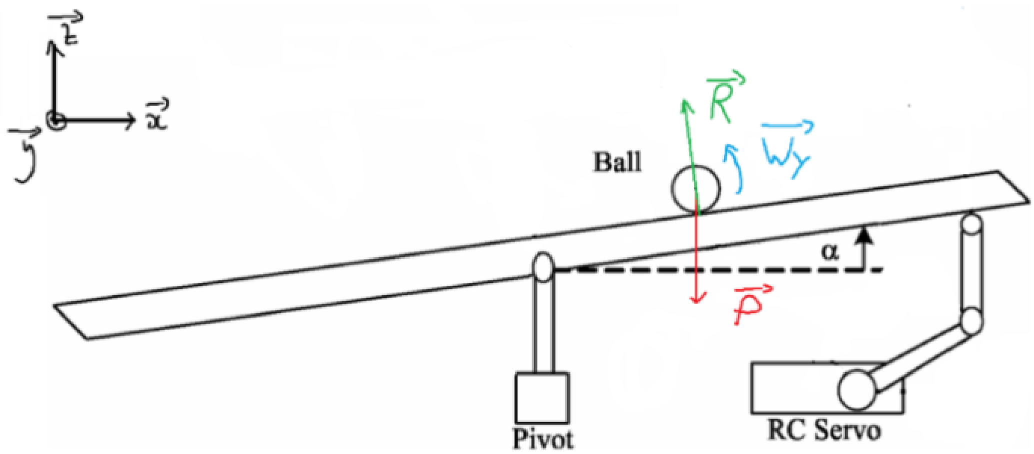


Figure 7: Representation of the forces in our physical modeling

This translation force is defined as: Relative to the x and y axis

$$F_{x,t} = m_b g \sin\alpha \quad F_{y,t} = m_b g \sin\theta$$

And the rotational force is

$$F_{r,t} = \frac{\tau}{r_b} = \frac{I_b \ddot{\alpha}}{r_b} = \frac{I_b \ddot{x}_b}{r_b^2}$$

with

$$\tau = I_b \ddot{\alpha}, \quad \alpha = \frac{\ddot{x}_b}{r_b}, \quad I_b = \frac{2}{5} m_b r_b^2$$

The total force acting on the ball is the sum of the translation force and the inertia force:

$$\sum F = (F_{x,t} + F_{y,t}) + F_{r,t} = m_b \ddot{x}_b$$

Then the acceleration of the ball:

$$\ddot{x}_b = \frac{m_b g \sin\alpha\, r_b^2}{m_b r_b^2 + I_b} \quad \ddot{y}_b = \frac{m_b g\, r_b^2}{m_b r_b^2 + I_b}$$

Parameters:

- α: The table tilt angle on the y axis.
- θ: The table tilt angle on the x axis.
- m_b: Mass of the ball with a diameter of 4 cm and a weight of 2.7 g.
- g: Acceleration due to gravity.

- $\ddot{\alpha}$: The angular acceleration.
- I_b: Moment of inertia of the ball.
- r_b: Ball radius.
- τ: The couple.

5.2.2 Transfer Function Analysis

Applying the Laplace transform to the equation of motion:

$$\ddot{x}_b = \frac{m_b g \sin\alpha\, r_b^2}{m_b r_b^2 + I_b}$$

$$\alpha = \frac{d}{L}\beta_y, \quad \theta = \frac{d}{L}\beta_x$$

Definition of Parameters

- β_y: The angle of inclination of the servo motor's base relative to the servo motor along the Y-axis.
- β_x: The angle of inclination of the servo motor's base relative to the servo motor along the X-axis.

$$\ddot{x}_b\left(m_b + \frac{I_b}{r_b^2}\right) = -m_b g \beta_y \frac{d}{L}$$

$$S^2 X(s)\left(m_b + \frac{I_b}{r_b^2}\right) = -m_b g \beta_y \frac{d}{L}$$

Final Transfer Functions

The final Transfer Functions for the X-axis:

$$T_x(s) = \frac{X(s)}{\beta_y(s)} = \frac{-m_b g r^2}{s^2\left(m_b + \frac{I_b}{r_b^2}\right) d}$$

For the Y-axis:

$$T_y(s) = \frac{Y(s)}{\beta_x(s)} = \frac{-m_b g r^2}{s^2 \left(m_b + \frac{I_b}{r_b^2}\right) d}$$

System Constants

$$T_x(s) = \frac{K_b}{s^2} \quad \Leftrightarrow \quad K_b = -\frac{m_b g r l}{\left(m_b + \frac{I_b}{r_b^2}\right) d}$$

5.2.3 Calculates PID constants

The system's transfer function is obtained using the Laplace transformation:

$$T_x(s) = \frac{X(s)}{\beta_x(s)} = \frac{K_b}{s^2}$$

where

$$K_b = -\frac{m_b g r l}{\left(m_b + \frac{I_b}{r_b^2}\right) d}$$

Substituting the values:

$$K_b = -\frac{0.0027 \times 9.81 \times 0.3}{\left(0.0027 + \frac{4.32 \times 10^{-7}}{(0.02)^2}\right) \times 0.05} = -42.08\,\mathrm{rad}^{-1} \cdot \mathrm{s}^{-2}$$

The gain K_b is negative due to the physical dynamics of the ball and plate system. When the plate is tilted in one direction (e.g., a positive angle $\beta_x > 0$), the ball moves in the opposite direction, rolling "downhill." This inverse relationship between the tilt and the ball's movement is captured by the negative sign in K_b's expression:

$$K_b = -\frac{m_b g r l}{\left(m_b + \frac{I_b}{r_b^2}\right) d}$$

In simple terms, if the plate tilts in one direction, the ball "moves away" in the opposite direction. This negative sign reflects this inverse relationship and is consistent with the natural dynamics of the system.

5.2.4 Relationship between the plate angle and the servo motor displacement d

The inclination angle α of a platform is related to the displacement d of a servomotor lever by the geometric relationship:

$$\alpha = \frac{d}{L}$$

Parameters:

- α: Inclination angle of the platform at time t.
- d: Displacement of the servo motor lever at time t.
- L: Length of the platform.

e. Servo motor speed

In this part, the speed of the motor is calculated:

$$\alpha = \beta \times \frac{d}{L}$$

so the tilt speed of the tray:

$$\dot{\alpha} = \frac{d\beta}{dt} = \frac{\dot{d}}{L}$$

so:

$$\ddot{\alpha} = \omega(t) \times \frac{d}{L}$$

$$\omega(t) = \ddot{\alpha} \times \frac{d}{L}$$

Parameters:

- $\omega(t)$: Angular speed of the servomotor.
- β: Rotation angle of the servomotor.

Motor speed and torque calculations based on ball load

$$\tau = I_b \ddot{\alpha}$$

Moment of inertia of the ball

$$I_b = \frac{2}{5} m_b r_b^2$$

Numerical Application

$$I_b = 4.32 \times 10^{-7}\,\mathrm{kg/m^2}$$

Parameters:

- $m_b = 0.0027$ kg
- $r_b = 2$ cm

Moment of Inertia of the Tray

$$I_p = \frac{2}{5} m_p r_p^2$$

Numerical Application

$$I_p = 0.0045\,\text{kg/m}^2$$

Parameters:

- $m_p = 0.6$ kg
- $r_p = 18.5$ cm

Torque of the Motor

$$I_{\text{total}} = I_p \times I_b$$

so

$$\tau = I_b \ddot{\alpha} = 5.5\,\text{kg} \cdot \text{cm}$$

Conclusion

Through this physical modeling, a deeper understanding of the dynamics between the ball and the tilting plate was achieved. The analysis considered both translational and rotational forces acting on the ball, as well as how servo motor movement influences the platform's angle and, in turn, the ball's motion. By deriving the transfer functions and calculating the system's inertia and torque, it became possible to tune a PID controller effectively and design a system that reacts precisely and predictably. This mathematical approach proved essential to creating a stable and responsive ball-balancing mechanism.

5.3 PID Control

Introduction Dynamic systems must therefore be controlled for stability and performance. The ball-and-plate system uses the PID controller to keep the ball at the chosen desired target by compensating for disturbances. The PID makes the plate tilt at different angles in real time using the proportional, integral, and derivative actions to eliminate the system errors and disturbance effects. Such a control procedure is fundamental in achieving accuracy, good structure dynamics, and the capability to recover from disturbances, making it the most appropriate for this application.This section provides information on the aims, constraints, technical description, testing procedure, conclusions, and risk management regarding the PID controller.

5.3.1 Objectives and Constraints

Objectives:
The PID controller's primary objectives are:

- Stabilize the ball at a target position.

- Minimize settling time (ts 2 s).

- Limit overshoot to less than 10

- Ensure robustness against disturbances and noise.

Constraints:

- Real-time computation: The PID calculations must be fast enough to maintain system stability.

- Hardware limitations: The motors must support the required angles and speeds.

5.3.2 Functional Diagram and Usage Scenario

The functional diagram below illustrates the closed-loop control system, highlighting the role of the PID controller.

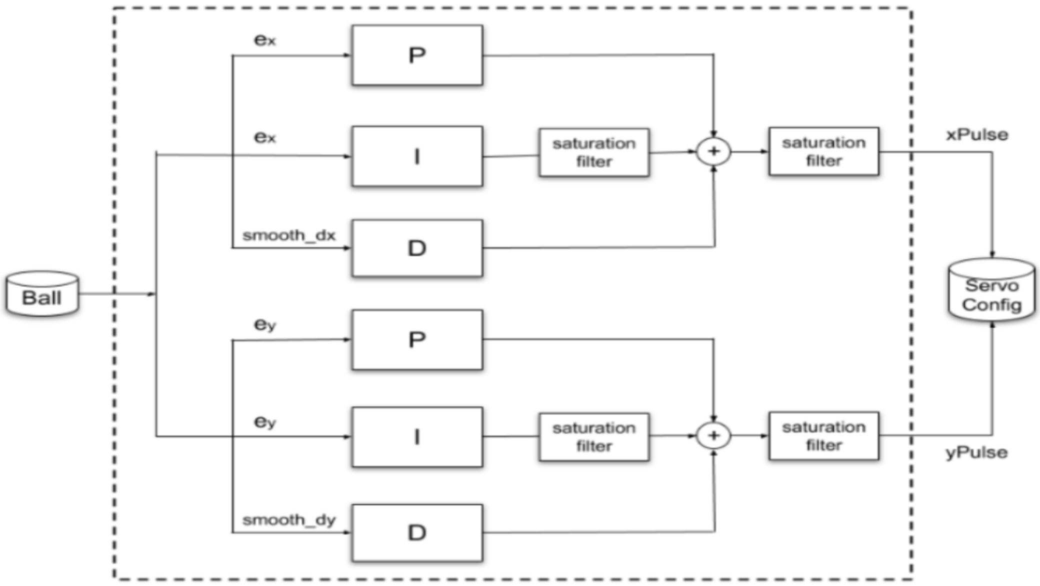


Figure 8: Schematic representation of the PID modeling

Two PID controllers operate independently to control the movement of the ball in the X and Y axes of the system. Such independence allows each controller to precisely adjust its own tilt angles of the plate so as to steer the ball in two dimensions. Each axis is handled independently, thus ensuring smooth, accurate, decoupled control, which simplifies the design and ensures stability in the presence of disturbance in one direction.

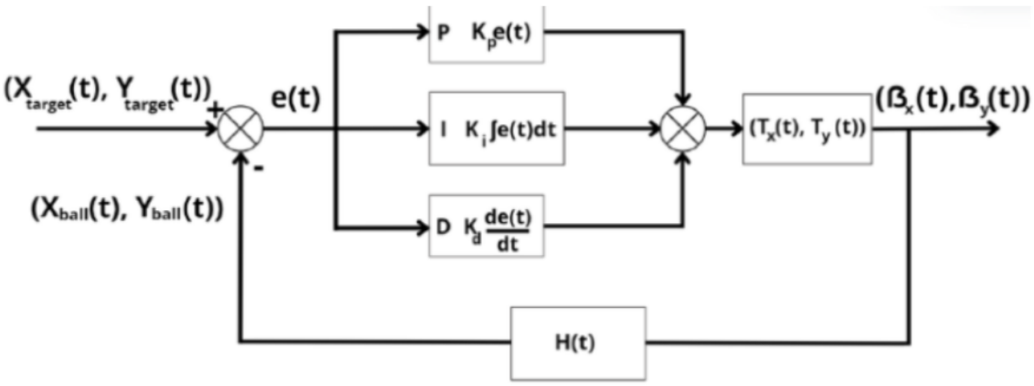


Figure 9: Closed loop control system diagram

The diagram shows the closed-loop control system for the ball-and-plate system. The target positions (X target,Y target) are, thus, set as the points. The error value is obtained from the difference between the measured positions (X measured,Y measured) and the target positions. This error is fed into the PID controller, which computes the commands (α, β) representing the tilt angles of the plate. These commands modify the incline of the plate, and the ball will roll toward the target positions due to the actions of the system dynamics.A camera H(t) acquires the real-time position of the ball, passing it to the vision module, which computes the measured coordinates. This feedback effectively becomes an inner loop controlling for error correction and stability in the ball's actual motion. The feedback loop makes the control process of the system precise and very reactive.

5.3.3 Technical Description and Testing

PID Design:

The PID controller is defined by its transfer function:

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

PID Gain Calculations:

The PID gains are calculated based on the performance specifications:

- Settling time: $t_s \approx 2$ s
- Maximum overshoot: $< 10\%$
- Damping ratio: $\zeta = 0.707$
- Natural frequency: $\omega_n = 4$ rad/s
- Additional real pole: $a = 4$

The formulas for the PID gains are:

$$K_d = \frac{a + 2\zeta\omega_n}{K_b}, \quad K_p = \frac{2a\zeta\omega_n + \omega_n^2}{K_b}, \quad K_i = \frac{a\omega_n^2}{K_b}$$

Substituting the values:

$$K_d = \frac{4 + 2 \cdot 0.707 \cdot 4}{-42.08} = -0.23$$

$$K_p = \frac{2 \cdot 4 \cdot 0.707 \cdot 4 + 4^2}{-42.08} = -0.92$$

$$K_i = \frac{4 \cdot 4^2}{-42.08} = -1.52$$

Stability Analysis

Characteristic Equation

The closed-loop system's characteristic equation is given by:

$$s^3 + K_b K_d s^2 + K_b K_p s + K_b K_i = 0$$

Substituting the values:

$$s^3 + 9.68s^2 + 38.71s + 63.96 = 0$$

Routh-Hurwitz Table

The Routh-Hurwitz criterion is a tool that determines how stable a system is by examining the signs of the first column of the Routh table. Such examination gives surety that all roots of the characteristic equation have negative real parts, a required condition for the stability of a system.

The Routh table is constructed as follows:

$$\begin{array}{c|cc} s^3 & 1 & 38.71 \\ s^2 & 9.68 & 63.96 \\ s^1 & \frac{9.68 \cdot 38.71 - 63.96}{9.68} = 238.47 & 0 \\ s^0 & 63.96 & 0 \end{array}$$

In our system, the coefficients in the first column of the Routh table $(1, 9.68, 238.47, 63.96)$ are all positive, confirming stability. Thus, the system is stable according to the Routh-Hurwitz criterion.

Implementation

The PID controller was implemented in Python as follows:

- **Error computation**: Calculate the difference between the target and actual positions.
- **PID corrections**: Compute the plate angles (θ_x, θ_y) based on K_p, K_i, and K_d.
- **Feedback loop**: Continuously update the ball's position and adjust the plate accordingly.

Tests and Results

The system was tested in the following scenarios:

- Stabilization of the ball at the target position.
- Introduction of perturbations to observe the recovery of the controller.

Results:

- **Stabilization time**: ≈ 2 s.
- **Overshoot**: ($< 10\%$), confirming that the damping $\zeta = 0.707$ ensures system stability.
- **Stability**: The ball remained stable at the target position after the perturbations.

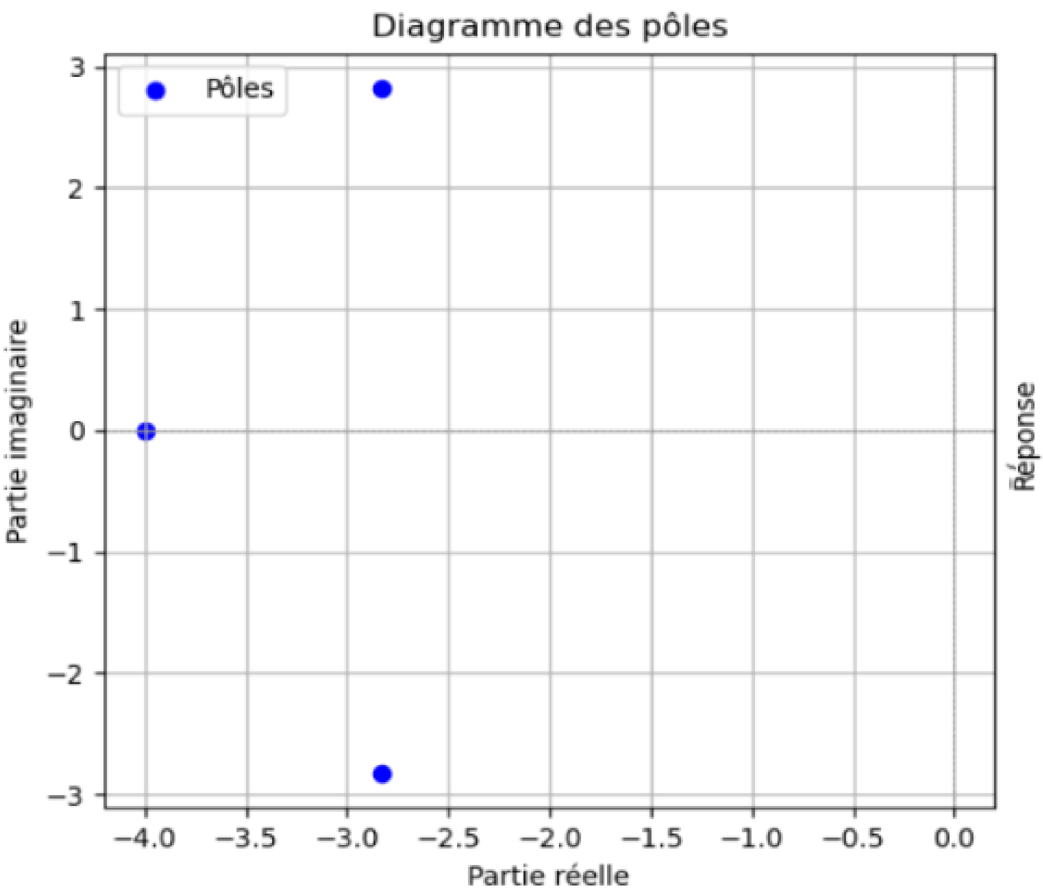


Figure 10: Pole diagram

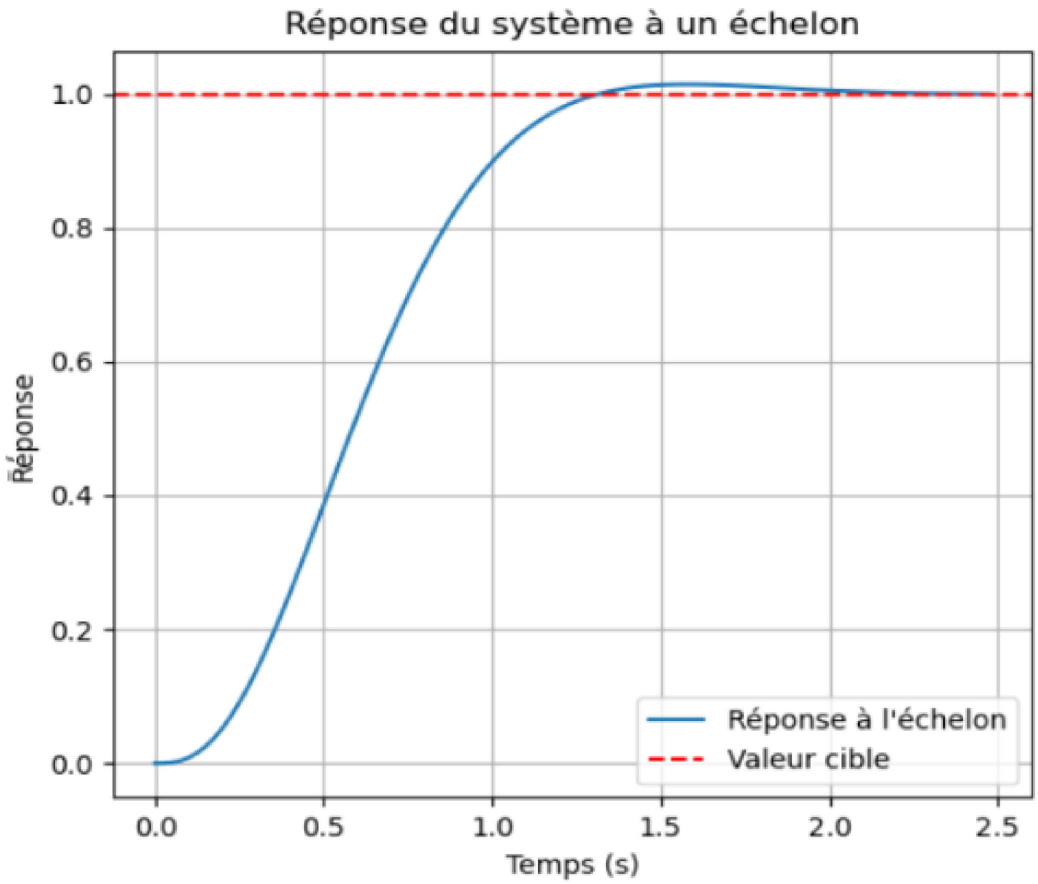


Figure 11: The step response

This diagram shows the poles of the closed-loop system. The dominant poles are located in the left part of the complex plane (negative real part), which ensures that the system is stable. The system reaches the setpoint in about 2 seconds, respecting the settling time specification. The overshoot is minimal and complies with the set limit ($< 10\%$), confirming that the damping $\zeta = 0.707$ has been well chosen. A gradual rise is visible, without marked oscillations, indicating a good combination of PID parameters.

Risk Analysis

The following risks were identified and mitigated:

- **Gain miscalculation**: This could lead to instability. Rigorous testing and simulations were used to avoid this.
- **Hardware limitations**: Components were selected based on real-time performance requirements.

Conclusion

The design of the PID controller for the ball-and-plate system was done painstakingly, using the transfer function and dynamic analysis of the system. Settling time ($t_s \approx 2$ s) and overshoot ($< 10\%$) were considered performance specifications used to determine the PID gains, with the Routh-Hurwitz criterion applied for checking the stability of the system. The implementation in Python showed good performance in stabilization and recovery from perturbations, thus attesting to the reliability of the design.

6 Technical choices

To take a step back and compare our technical choices, it is important to evaluate the different options considered throughout the project. For example, during the selection of the motors, we compared several models in terms of torque, speed, and compatibility with our system. The final choice, oriented towards a motor offering a good compromise between power and precision, proved crucial to ensure the responsiveness, stability, and precision that are important for our project. Similarly, for the analysis of the ball's position, we chose to use a high-frequency camera module V2, which proved to be more flexible and suited to our need for real-time tracking. Similarly, for data processing and real-time control, we opted for a Raspberry Pi 5. This choice was made after comparing its performance with other microcontrollers. The Raspberry Pi 5 stood out for its superior computing power, necessary for handling image processing and PID algorithms.
The technical choices are presented in tabular form, allowing for a clear comparison with other available options. This approach facilitates the evaluation of the features, while highlighting the advantages and disadvantages of each choice to guide the final decision.

Raspberry Pi

Criterion	**Raspberry Pi 4**	**Raspberry Pi 5 (choice)**	**Justification**
Processor	ARM Cortex-A72, Quad Core, 1.5GHz	ARM Cortex-A76, Quad Core, 2.4GHz	Sufficient computational power for PID and image processing.
RAM	2 GB, 4 GB, 8 GB	Up to 8 GB	Enough memory to store and process real-time images.
GPIO Ports	40-pins	40 pins	Allows connection of multiple components (motors, camera, etc.).
Connectivity	Gigabit Ethernet, Wi-Fi, Bluetooth	USB, HDMI, Ethernet, Wi-Fi	Provides flexibility for development and testing.
Operating System	Raspberry Pi OS	Linux compatible (Raspbian OS)	Broad compatibility with libraries for PID and image processing.

Table 2: Comparison between Raspberry Pi 4 and Raspberry Pi 5

Pi Camera V2

Criterion	Value / Specification	Justification
Resolution	8 **megapixels**	Provides precise detection of the ball's position.
Frame Rate	Up to 120 fps (at lower resolution)	High frame rate to minimize capture latency.
Interface	CSI (Camera Serial Interface)	Compatible with Raspberry Pi for direct, fast connection.
Latency	Low	Essential for real-time adjustments in ball tracking.
Dimensions	25mm x 23mm x 9mm	Compact size for easy integration on the platform.

Table 3: Specifications of the Pi Camera V2

The MG90S Servo

Criterion	FS5115M	MG90S (choice)	Justification
Torque	10 kg·cm à 4.8V	3 kg·cm at 4.8V	Sufficient force to handle tilting the platform and stabilizing the ball's motion.
Rotation Speed	0.16 s/60°	0.2 seconds per 60 degrees	Allows quick adjustments for real-time control of the platform tilt.
Operating Voltage	4.8 à 6V	4.8V to 6.0V	Compatible with standard power supplies used in control systems like Raspberry Pi or similar setups.

Table 4: Comparison between FS5115M and MG90S Servo Motors

7 Final System Integration

The final integration of the system was an important step to complete the project. We assembled all the components: the circular platform, the MG90S servo motors, the Pi V2.1 camera, and the Raspberry Pi 5. The Raspberry Pi controls everything: it captures images, processes the data, and calculates how to move the platform. The camera was placed above the platform using a strong support to keep it stable and give a clear view. The servo motors were installed under the platform with flexible arms to allow the platform to tilt in two directions.

Replacement of Motor Supports

The mechanical structure—especially the platform and its supports—was originally built by students from previous years. During early testing, it became clear that the motor supports (the feet) were either damaged. As a result, the motors struggled to move the platform as intended. In some cases, the platform would get stuck or tilt unevenly.

To fix the problem, the old supports were replaced with newly designed, more durable ones. These reinforced parts provided better stability and helped the motors operate more smoothly. After this change, the platform moved more freely and reacted more accurately during control tests.

The images below highlight the difference between the damaged original parts and the improved replacements.

figure 1 : old support

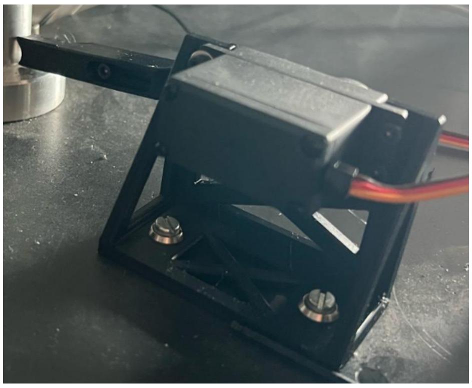


figure 2 : new support

Figure 12: Comparison between the old support support and the new 3D-printed one

Three new support parts were designed using Autodesk Fusion 360, carefully modeled to match the required dimensions with precision. The first linkage arm serves as the connection between the servo motor and the pivot joint. It is mounted directly onto the servo and houses two ball bearings to ensure smooth rotational movement. Although the bearings are relatively small, the pivot joint was designed to accommodate both, providing excellent rigidity and structural stability to the system.

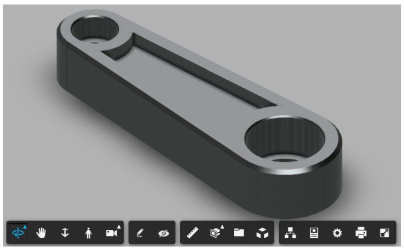


Figure 13: Motor arm

The second linkage arm is composed of two sliding components, allowing its length to be finely adjusted. This adjustable design makes it easy to calibrate the plate's neutral position with high precision. The two parts can slide into each other and are locked in place with a screw that passes through them.

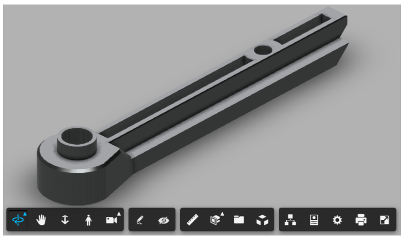


Figure 14: Tray arm 1

To ensure accurate assembly, the first part of the linkage arm includes a machined collar and a precision hole, both processed after 3D printing, to perfectly fit the bearing axle. The second part includes two magnet housings and a spherical cavity designed to hold the ball of the ball joint, allowing for flexible movement.

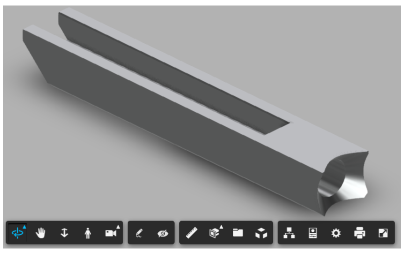


Figure 15: Tray arm 2

After 3D printing the parts, the system was reassembled, and this time, the motors were firmly secured and operated as intended. This improvement made it possible to continue the project with confidence, without further mechanical issues disrupting progress.

8 Hardware Setup and Control Mechanism

8.1 Component list

Component	Model	Function
Raspberry	Pi 5	Central unit for image processing, control loop, and GPIO signal output
Camera Module	Raspberry Pi Camera V2.1	Captures real-time video for ball detection
Servo Motor (x2)	MG996R	Controls the inclination of the platform on X and Y axes
Jumper Wires	female-to-femaleMale-to-female	Connections between Raspberry Pi GPIO pins and servos
Ball	ping pong ball	Tracked target for stabilization experiment
Tray	Wooden Circular	Surface on which the ball moves; mounted on two servo axes
Power Bank	5V, 2A USB Output	External power supply for the servos
Tray Support Structure	Plastic	Holds the servo motors in place and allows for tray tilt

Table 5: List of components used in the project

8.2 Hardware diagram

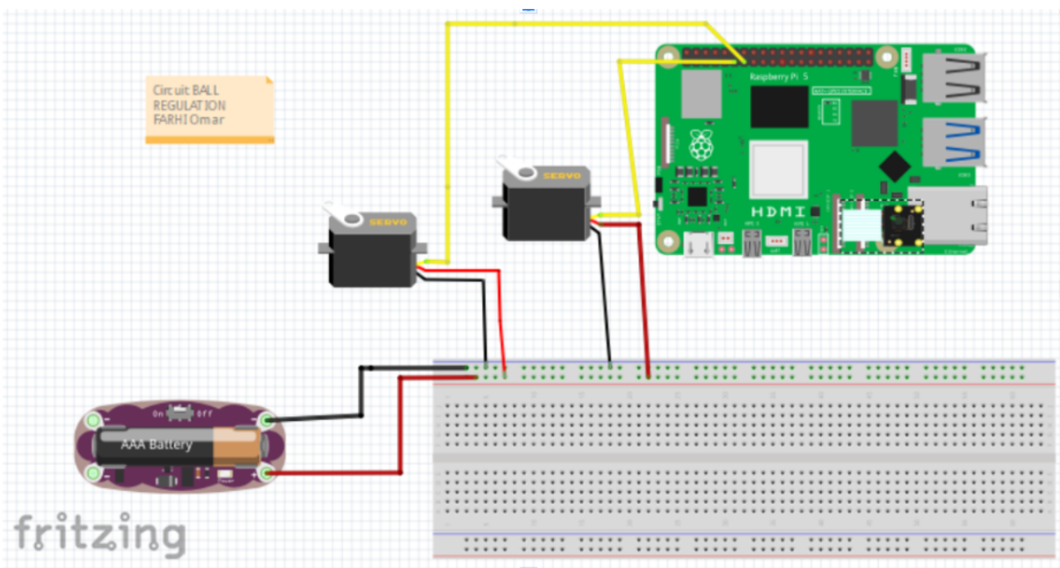


Figure 16: Project circuit

The circuit, designed using Fritzing, illustrates the hardware architecture of the ball tracking system, highlighting the connections between key components. At the core of the system is a Raspberry Pi, which serves as the main controller. A camera, connected to the Raspberry Pi via the CSI interface, captures real-time video to enable the detection of the ball and plate. Two servo motors are connected to the Espace pour éviter que ça touche l'en-têteRaspberry Pi's GPIO pins to enable precise tilt adjustments of the plate: one is linked to GPIO12 (X-axis) and the other to GPIO13 (Y-axis), each receiving PWM signals for accurate movement control. A breadboard is used to streamline the electrical connections. An AA battery, depicted as an external power source, supplies the necessary energy to the servomotors. This regulation circuit is crucial for stabilizing the voltage and preventing current fluctuations from the servo motors from affecting the Raspberry Pi, ensuring a reliable and consistent power supply. The connection wires, clearly shown, illustrate the links between the battery, breadboard, servo motors, and Raspberry Pi, using distinct colors for signals (yellow for PWM signals, red and black for power and ground).

9 Software Workflow

The first diagram (Figure 17) provides an overview, illustrating the main stages of the process. It condenses the operations into six key steps: Open Video Stream, Initialize Servo Communication, Initialize Data Structures, Acquisition, Control, and Actuation. In contrast, the second diagram (Figure 18) is far more granular, detailing every specific step of the process with a total of 20 step: The process starts by initializing the system: the camera is activated to capture a continuous video stream, providing the images needed for tracking. A connection is established with the servomotors via GPIO pins, enabling their control, while PID controllers (Proportional, Integral, Derivative) are configured to adjust the ball's position based on detected deviations. Detection variables, such as thresholds and coordinates, are set, and an initial FPS (frames per second) measurement is performed to evaluate baseline performance. The system then enters a loop: a new frame is extracted from the video stream, converted to grayscale to simplify processing, and a Gaussian blur is applied to reduce noise and enhance detection accuracy. The Hough Transform is used to identify the plate, followed by the ball within it. The ball's position is normalized relative to the plate's center, and the PID controller calculates the necessary adjustments to recenter it. These corrections are limited to prevent abrupt servomotor movements, which then receive commands to adjust the plate's tilt. At each cycle, the system checks for a termination condition (for example, a key press): if not met, it returns to capturing a new frame; otherwise, it proceeds to stop the video stream, close any display windows, display performance statistics like the average FPS, and terminate the process.

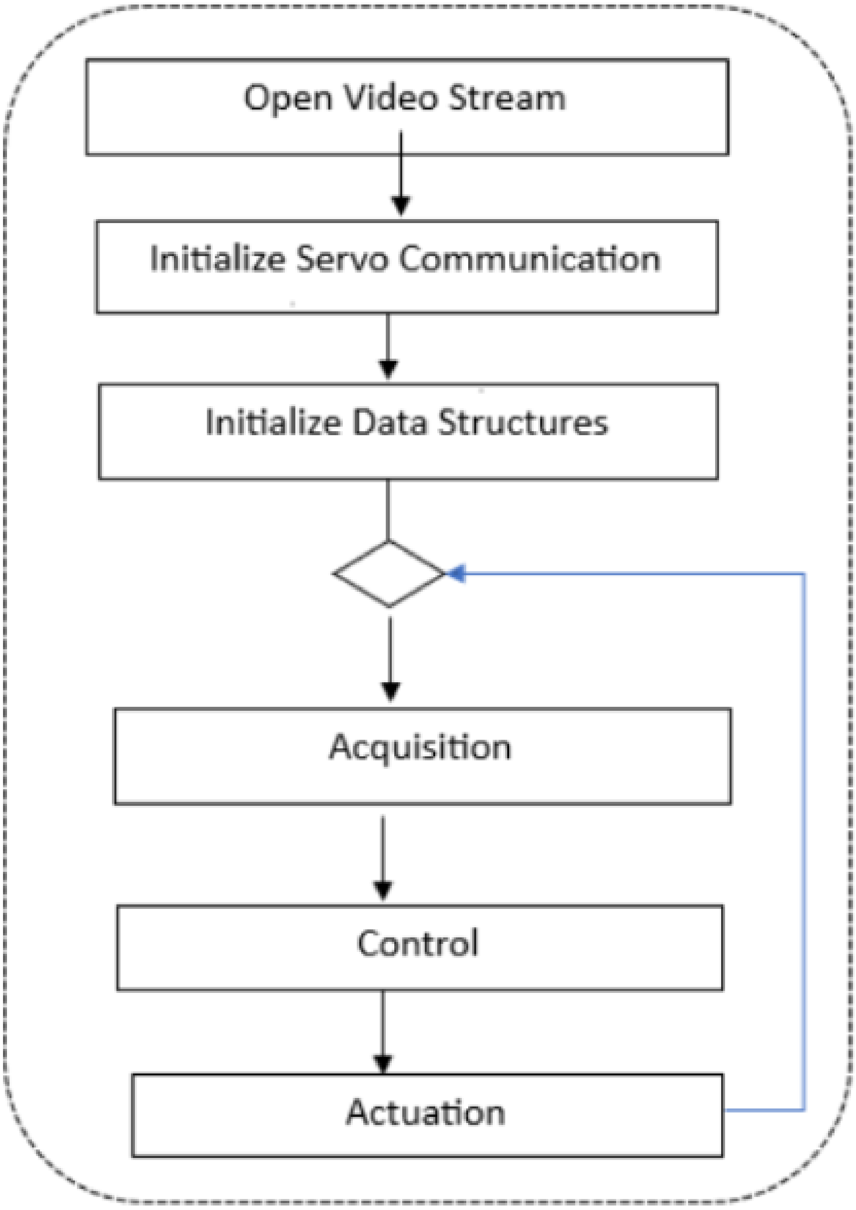


Figure 17: The six keys of the global code

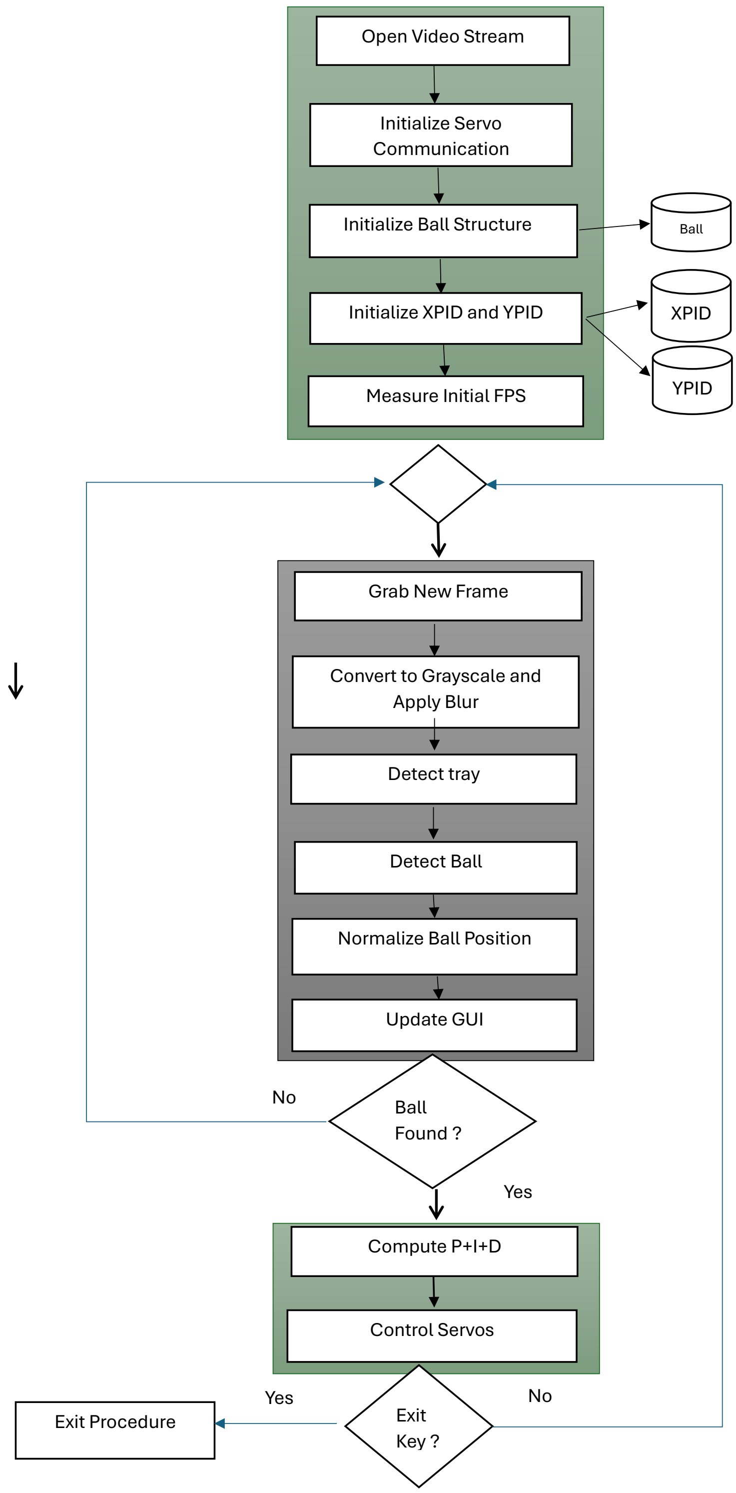


Figure 18: Flowchart of the Ball Stabilization System

10 System limitations and possible improvements

Identified Problem / Limitation	Probable Cause	Impact on the System	Proposed Solution
1. Poor detection of the plate when it tilts	Image distortion, variable lighting, overly sensitive algorithm	Incorrect detection of plate center → inaccurate ball position	Apply camera calibration, use ellipse/contour detection, or improve robustness
2. Instability or slow response of the motors	unstable PWM signal	Motors do not apply correct angles → ball fails to stabilize	Install pigpio, ensure GND is shared, use a stable 5V power supply
3. Inconsistent motor behavior despite correct PID	Misinterpreted PWM signal, servo saturation, incorrect PWM range	PID output is correct checked by simulation but motors respond incorrectly	Adjust PWM mapping (e.g. 5–10%), test PWM range independently with fixed values
4. Sensitivity to lighting conditions during detection	Uncalibrated camera, lighting variation, no brightness normalization	Ball or plate is poorly detected under certain lighting conditions	Add brightness correction or uniform LED lighting, apply image normalization
5. Transfer function not validated	System not modeled or transfer function not verified	PID tuning not based on real system behavior	Recalculate and verify the transfer function with step response and parameter fitting

Table 6: Summary of Identified Problems, Causes, Impacts, and Solutions

11 Risk Analysis Table

Risk Type	Description	Severity	Probability	Impact	Preventive Measures
Technical					
Motor Failure	Motor stops or causes uncontrolled tilt.	High	Medium	Instability of the ball and platform.	Add failure sensors, including an emergency stop.
Image Processing Error	Incorrect ball detection due to algorithm issues.	Medium	Medium	Impacts ball movement and system accuracy.	Use redundant algorithms or a manual correction mode.
Communication Problem	Delayed data between camera and motors.	High	Medium	Slower system response, instability.	Optimize protocols, reduce latency, and check wiring.
Motor Response Delay	Motors are slow to react to commands.	High	Medium	Instability, poor performance.	Use faster motors, adjust PID, and test in real conditions.
Safety					
User Interference	Physical interference disrupts ball movement.	Medium	Medium	System errors, unpredictable movement.	Add safety mode to pause the system during interference.
Camera Obstruction	Camera blocked by user or dust.	High	Medium	Detection failure, interruptions.	Add alerts or auto-pause for obstructions.
Environmental					
Lighting Conditions	Poor lighting affects ball detection.	Medium	Medium	Difficult tracking, less accuracy.	Use consistent lighting or adapt detection algorithms.
Financial					
Budget Overrun	Component or testing costs exceed estimates.	Medium	Low	Limits resources for other tasks.	Track expenses with a buffer in the initial budget.
Human					
Human Error	User actions disrupt ball movement.	Medium	Medium	System disturbance, unpredictable behavior.	Add tolerance mechanisms or alerts for unusual actions.
Durability					
Component Wear	Mechanical parts degrade over time.	Medium	Medium	Reduced performance, increased failures.	Regular maintenance, use high-quality components.

Table 7: Risk Analysis Table

12 Conclusion

This report established a structured methodology for designing a ball-position control system on a tiltable platform, demonstrating the feasibility of combining physical modeling, real-time image processing, and PID control in the implementation of the system. The configuration, through dynamic adjustments, effectively stabilizes the ball in the center of the platform, showcasing the prospects of the mechanics-electronics-computer science integration to solve the timeliest of problems. However, choices on certain models used, including the Raspberry Pi 5 and Pi V2.1 camera, were made after careful consideration of their capabilities to meet the expectations from the project and needed speed of the program, accurate image processing, and consequent drive combination for real-time effectiveness.

There is burgeoning scope for improvement, including more focus on providing optimal lighting conditions to avoid shadows for better ball detection. If a solution can be implemented, it would adequately elevate the performance of the system in terms of precision and robustness, particularly so in lighting-deficient conditions. Tuning of the PID controller parameters can also be sought in the near future for a more responsive and more stable control system to yield a smooth and easy operation.

The project, this, is not only a proof of concept for the envisaged application but also a solid platform on which future developments can be built. It raises the prospects of more advanced control strategies, adding more sensors, and scaling to more involved scenarios. The project stands up to introducing a great educational tool for practical training in real-time control systems, computer vision, and transdisciplinary problem-solving. Addressing the aforementioned suggested improvements and successfully transitioning from theory to practical implementation signals the technical aplomb and innovativeness of the team and the institutions involved.

Table of Annexes

Annex A.1: Capella - Mission Capabilities Blank Diagram

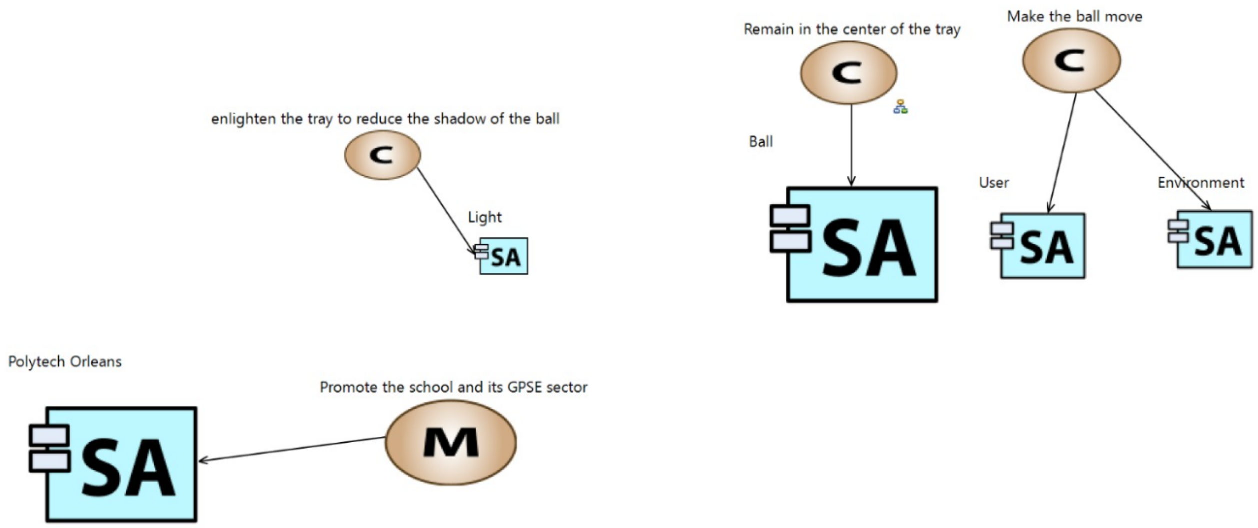


Diagram showing the system architecture using Capella.

Annex A.2: Capella - System Architecture Diagram

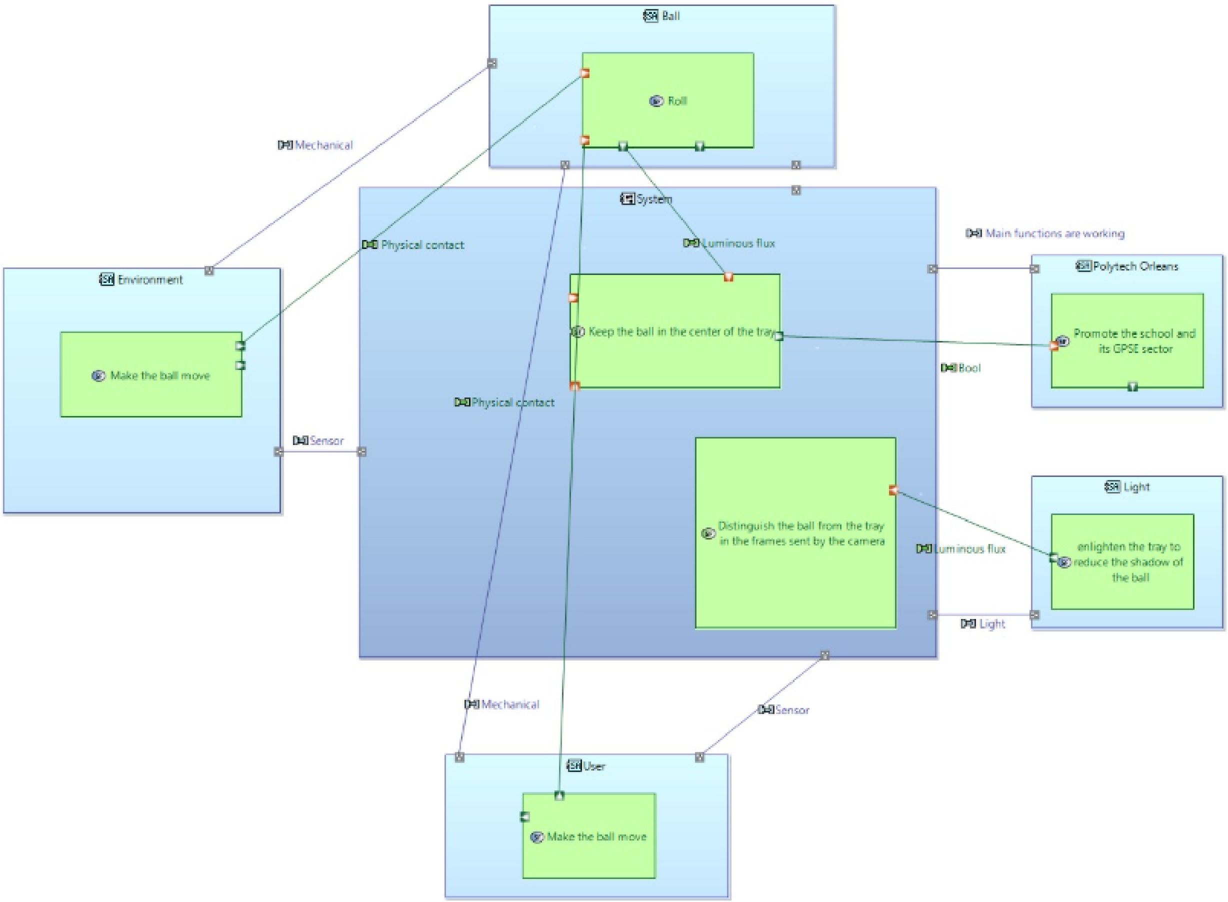


Diagram showing the system architecture using Capella.

Annex A.3: Capella - System Architecture Diagram

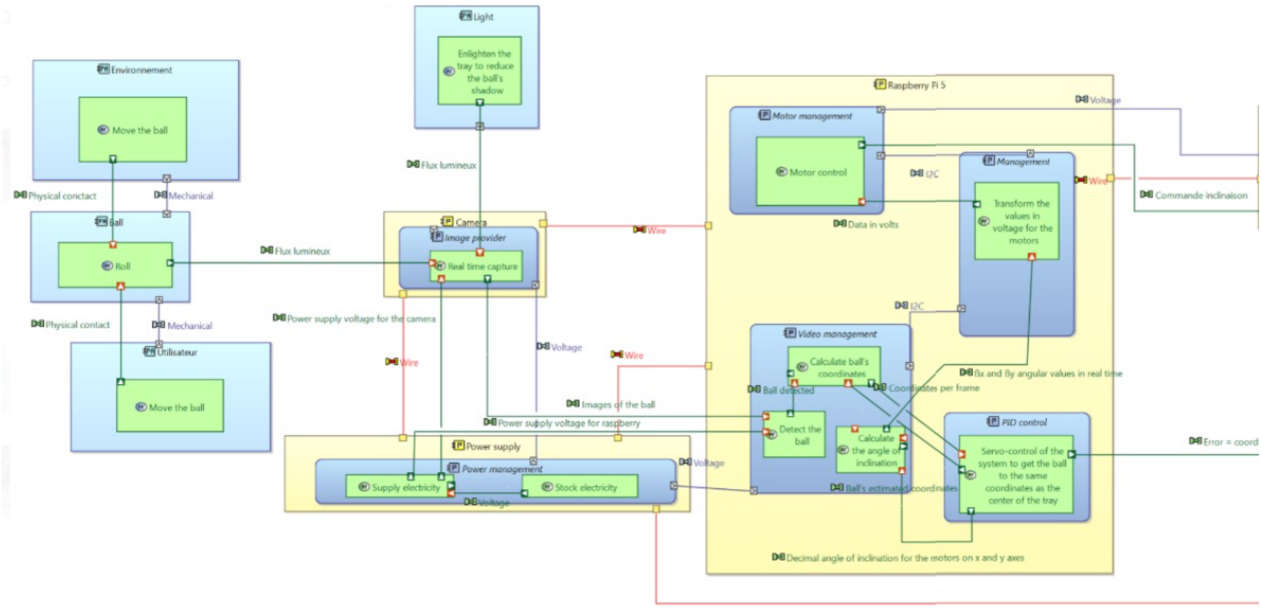


Diagram showing the system architecture using Capella.

Annex A.4: Capella - System Architecture Diagram

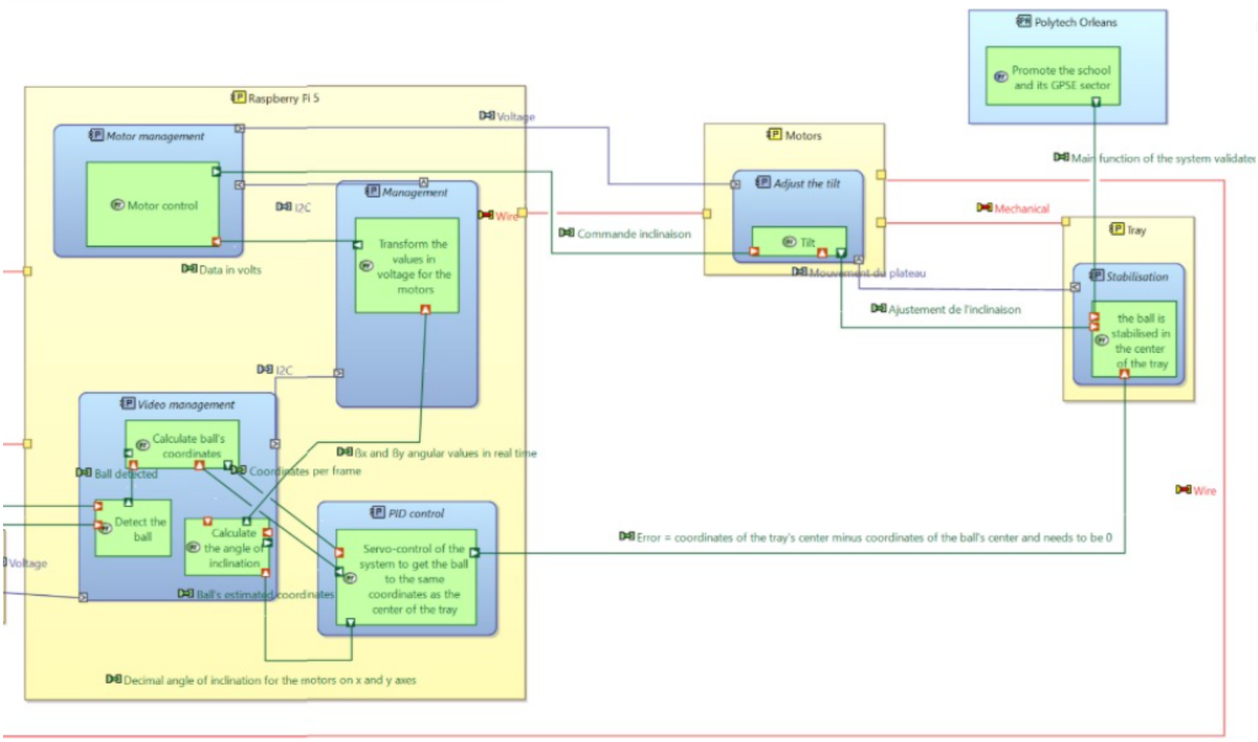


Diagram showing the system architecture using Capella.

Annex B: Details of Calculations

Routh-Hurwitz Table Construction

The characteristic equation is given as:

$$s^3 + 9.68s^2 + 38.71s + 63.96 = 0$$

The coefficients are:

$$a_3 = 1, \quad a_2 = 9.68, \quad a_1 = 38.71, \quad a_0 = 63.96$$

Filling the Table

First Row: The first two coefficients of the characteristic equation:

$$a_3 = 1, \quad a_1 = 38.71$$

Second Row: The next two coefficients:

$$a_2 = 9.68, \quad a_0 = 63.96$$

Third Row: Compute the element using the formula:

$$\text{Element} = \frac{(a_2 \cdot a_1) - (a_0 \cdot a_3)}{a_2}$$

Substituting:

$$\frac{(9.68 \cdot 38.71 - 63.96)}{9.68} = \frac{375.77 - 63.96}{9.68} = \frac{311.81}{9.68} \approx 238.47$$

Completed Routh Table

The final Routh table is:

$$
\begin{array}{c|cc}
s^3 & 1 & 38.71 \\
s^2 & 9.68 & 63.96 \\
s^1 & 238.47 & 0 \\
s^0 & 63.96 & 0
\end{array}
$$

For the system to be stable, all elements in the first column of the table must be positive:

$$1,\ 9.68,\ 238.47,\ 63.96$$

Since all values are positive, the system is stable according to the Routh-Hurwitz criterion.

Annex C: Competence Evaluation

Function performed	Description	Salma NAHROU	Omar FARHI	Jérôme DUFFEL
Programming	-Developing the control algorithms PID and Simulation.	5%	90%	5%
	-Global code.	55%	45%	_
	-Code image processing	5%	5%	90%
Mechanical study	-Performing technical solutions.	90%	5%	5%
	-PID calcul.	10%	90%	_
Image processing	-Code development.	50%	50%	_
Implementation	-Hardware Diagram.	20%	80%	_
	-Software Workflow.	90%	10%	_
	-Diagram of Ball Detection.	70%	30%	_
	-Capella	5%	5%	90%

Diagram showing the mission-level capabilities identified using Capella.

Annex D: Auto-évaluation

Omar Farhi

During this project, I learned many useful skills. I worked on real-time image processing to detect the ball's position and used a PID controller to control the platform. I now understand how PID works and how each gain (P, I, D) has a different role in controlling the system. I saw how changing the gains can make the system faster or more stable.

It was also my first time working with a Raspberry Pi. I learned how to use it, how to connect components, and how to configure it to control the motors.

This project helped me improve my programming skills in Python and also taught me how to analyze and simulate different situations. I understood better how hardware and software can work together.

I also did some mechanical calculations and participated in system analysis using diagrams like FAST. I worked with my teammates, which helped me improve my communication and teamwork

I really enjoyed the project even the challenges . I feel more confident now and more prepared as a future engineer.

Salma Nahrou

During the ball control project, I had the chance to sharpen my technical skills and pick up some new knowledge along the way. In the second phase, I dove into developing real-time image processing code with the OpenCV library in Python, focusing on detecting and tracking a ball's trajectory. This part of the project also me to the Raspberry Pi electronics board, where I learned how to set it up for image capture and connect it with another components

The second aspect of the project was implementing a PID controller to stabilize our system. This meant fine-tuning it under real-world conditions, taking into account things like servo motor response time, system latency, and the limited range of motion. I also took the initiative to create a physical model of the system to identify crucial parameters for our technical decisions and establish the transfer function for our PID.

I also worked on the global code that brought together image processing, motor control, and the PID system. This integration was essential for ensuring good communication between all components. The project was both challenging and rewarding, regrouped coding, electronics, and control systems.

This experience highlighted the importance of thorough preparation before a land to develop. I contributed to the FAST diagrams, the risk analysis, defining technical specifications, and setting up a Gantt chart. These planning helped our team to manage tasks and meet deadlines and have an overall view of the progress of the project during the 2 semesters. I also learned that paying attention to all the details and constraints from the start is vital for completing a project successfully.

All in all, the project deepened my understanding of teamwork, information sharing, and taking responsibility within a group. While it was a tough experience, it helped me grow in these areas and taught me how to tackle and solve real-world problems effectively.

Jérôme DUFEIL

In this project I have a special status because in addition to having to carry out my tasks I also have the important role of project manager. I did not only learn new skills, whether methodologically or technically, as well as reinforced other skills, but I also discovered the weight of responsibility.

At the beginning of the project, I realized that my approach as a project manager was wrong, and this led to some confusionin the calculations for the motors. As a result, we encountered significant delays in studying the motors, which ultimately prevented us from ordering them on time and caused a setback in the project timeline. Reflecting on this experience, I understood the importance of setting clear deadlines, regularly checking progress, and maintaining team motivation to ensure the work stays on track. To help recover from these delays, I dedicated time to assisting my team with the theoretical study, refining constraints, and performing calculations more efficiently. This experience gave me valuable insights into the responsibilities of a project manager and allowed me to develop a better understanding of leadership challenges. It also highlighted areas where I need to grow, as I realized I currently lack the skills and experience necessary to manage a team effectively. Nonetheless, this role gave me a glimpse into the demands of leadership and motivated me to improve in this area.

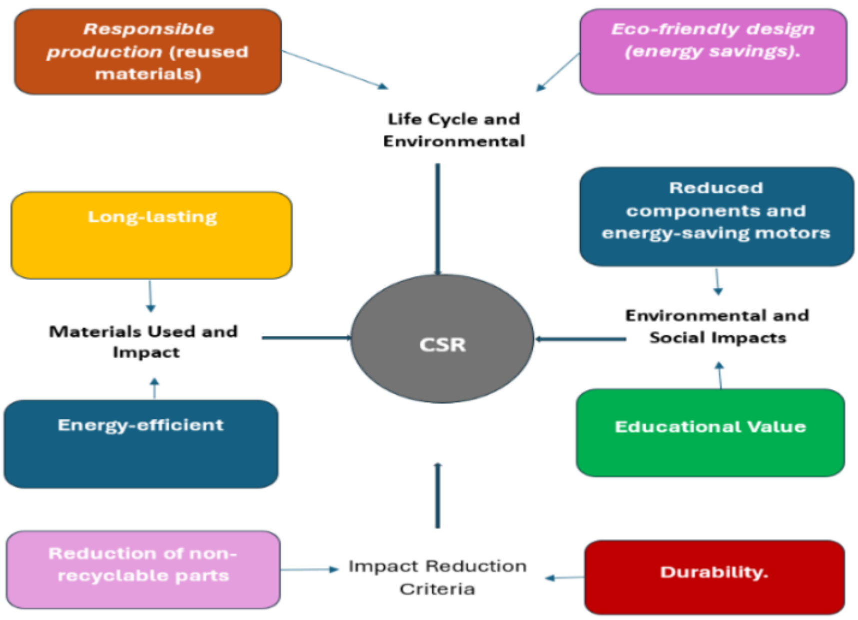
As for the tasks I realised, this project helped me to improve and develop new hard skills such as mastering Capella or image processing which deepened my knowledge in Python. Furthermore, I dove back into physical modeling and mathematics with Hough transform for instance. In addition, I had fun making all kinds of diagrams, as well as the poster which shaped my graphic skills. Moreover, I deepened my soft skills like project management, stress management, team management, my adaptability and my autonomy, except for project planning which I haven't laid hands on.

Although I will not be present in the second semester to keep working on this project to develop new skills, each more interesting than the other, or to conform to my already possessed skills, I really enjoyed working on this project which ultimately appealed to me whereas at the beginning it was more of a choice out of spite. While I know that I lack certain

Annex E: Corporate Social Responsibility (CSR)

This project is built with a strong emphasis on both environmental and social sustainability. Environmentally, the design prioritizes simplicity, using fewer components to reduce energy consumption and electronic waste over the system's *lifecycle*. For instance, energy-efficient motors are chosen to minimize power usage, and materials like durable, recyclable plastics for the camera housing and wood for the plate ensure that components can be reused or recycled, lessening the ecological impact. The inclusion of an optimized PID control algorithm further enhances efficiency by activating the motor only when necessary, resulting in reduced energy usage overall.

From a social perspective, the project serves as an educational tool, equipping students and researchers with practical knowledge in automation and programming. Its open-source framework encourages collaboration and knowledge sharing, making it an accessible platform for learning and improvement within the robotics community. By combining an efficient, durable design with a commitment to reducing waste, the project seeks to provide long-term value while supporting sustainable practices and fostering skill development.



CSR diagram showing the lifecycle and environmental/social sustainability aspects.

Annex F: Bibliography and Links

- Sciendo.com: A real time control system for balancing a ball on a platform with FPGA parallel implementation
- https://freddy.mudry.org/public/NotesApplications/NAPidAj_06.pdf
- https://automaticsolution.wordpress.com/2020/04/28/la-regulation-pid/comment-page-1/
- https://dspace.unmto.dz/items/0e2b4a70-a073-4277-bd8b-e48b78b7c042
- HSV: https://docs.wlip.org/fr/stable/docs/software/vision-processing/wpilibpi/image-thresholding.html
- Hough Transform: https://medium.com/@afrah02shaikh/perform-hough-transform-for-circle-detection-on-the-image-x-using-python-b3e7db95f18e

3D Models of Motor Support Parts (Autodesk Fusion 360 V.2.0.15299)

The following links provide access to the 3D models of the motor support parts designed using Autodesk Fusion 360:

- Model Link 1
- Model Link 2
- Model Link 3

Project Repository (Github)

The complete code, documentation, and branches of the project can be accessed via the following GitLab repository:

`https://github.com/salma231-d/ball-regulation/branches`